# 4.6 Astrophysics and cosmology

This topic covers the physical interpretation of astronomical observations, the formation and evolution of stars, and the history and future of the universe.

| Students will be assessed on their ability to: | Suggested experiments |
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| 126 use the expression $F = Gm_1m_2/r^2$ | |
| 127 derive and use the expression $g = -Gm/r^2$ for the gravitational field due to a point mass | |
| 128 recall similarities and differences between electric and gravitational fields | |
| 129 recognise and use the expression relating flux, luminosity and distance $F = L/4\pi d^2$ | application to standard candles |
| 130 explain how distances can be determined using trigonometric parallax and by measurements on radiation flux received from objects of known luminosity (standard candles) | |
| 131 recognise and use a simple Hertzsprung-Russell diagram to relate luminosity and temperature. Use this diagram to explain the life cycle of stars | |
| 132 recognise and use the expression $L = \sigma T^4 \times$ surface area, (for a sphere $L = 4\pi r^2\sigma T^4$) (Stefan-Boltzmann law) for black body radiators | |
| 133 recognise and use the expression: $\lambda_{max}T = 2.898 \times 10^{-3}$ m K (Wien's law) for black body radiators | |
| 134 recognise and use the expressions $z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for a source of electromagnetic radiation moving relative to an observer and $v = H_o d$ for objects at cosmological distances | |
| 135 be aware of the controversy over the age and ultimate fate of the universe associated with the value of the Hubble Constant and the possible existence of dark matter | |

| Students will be assessed on their ability to: | | Suggested experiments |
|---|---|---|
| 136 | explain the concept of nuclear binding energy, and recognise and use the expression $\Delta E = c^2\Delta m$ and use the non SI atomic mass unit (u) in calculations of nuclear mass (including mass deficit) and energy | |
| 137 | describe the processes of nuclear fusion and fission | |
| 138 | explain the mechanism of nuclear fusion and the need for high densities of matter and high temperatures to bring it about and maintain it | |

# Context-led approach (based on the Salters Horners Advanced Physics project)

The following section shows how the specification may be taught using a context-led approach.